

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Key Features:**
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- **Missing Data:** 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created `age_group` column by binning customer ages.
 - Created `purchase_frequency_days` column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected the Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL Workbench to answer key business questions. The database and all ten queries below are from `customer_behavior_sql_queries.sql`.

```
CREATE DATABASE customer_behavior;
USE customer_behavior;
```

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

gender	revenue
Female	75191

gender	revenue
Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

Total rows returned: 839

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

```
SELECT item_purchased,
       ROUND(AVG(review_rating), 2) AS `Average Product Rating`
FROM customer
GROUP BY item_purchased
ORDER BY `Average Product Rating` DESC
LIMIT 5;
```

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

```
SELECT shipping_type,
       ROUND(AVG(purchase_amount), 2) AS avg_purchase_amount
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

shipping_type	avg_purchase_amount
Standard	58.46
Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

```
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
```

```

        ROUND(AVG(purchase_amount), 2) AS avg_spend,
        ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;

```

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

```

SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes'
                              THEN 1 ELSE 0 END) / COUNT(*), 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;

```

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

```

WITH customer_type AS (
  SELECT customer_id, previous_purchases,
         CASE
           WHEN previous_purchases = 1 THEN 'New'
           WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
           ELSE 'Loyal'
         END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(*) AS `Number of Customers`
FROM customer_type
GROUP BY customer_segment;

```

customer_segment	Number of Customers
Loyal	3116
New	83
Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

```

WITH item_counts AS (
  SELECT category,
         item_purchased,
         COUNT(customer_id) AS total_orders,
         ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS
item_rank
  FROM customer
  GROUP BY category, item_purchased
)

```

```
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
```

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with more than 5 previous purchases are more likely to subscribe.

```
SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

subscription_status	repeat_buyers
No	2518
Yes	958

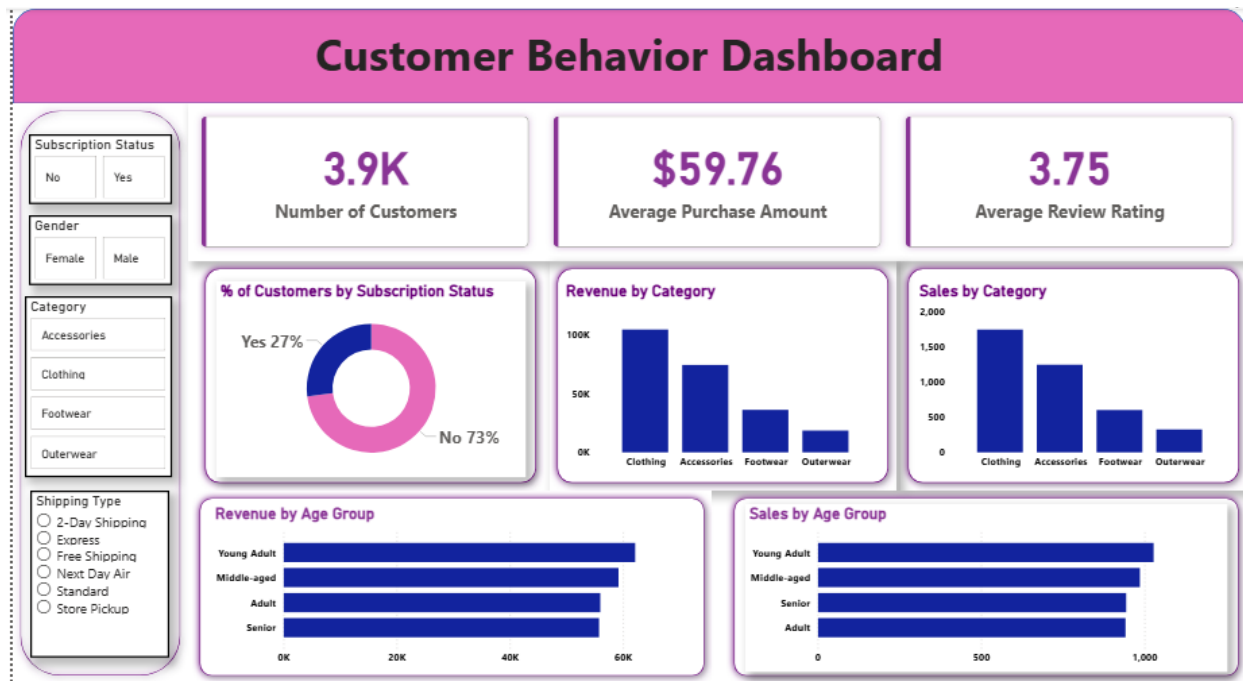
10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

```
SELECT age_group,
       SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.